AI Browser Extension for Teen Safety

Protecting young users from harmful online content through intelligent, real-time browser-level intervention

THE CHALLENGE

Underage Access to Harmful Content

The Problem

Minors routinely access pornographic, gambling, betting, and dark-web content through standard browsers, leading to psychological harm, addiction, financial loss, and broader social issues affecting teenagers, families, and educational institutions.

Why Current Solutions Fall Short

Traditional parental controls require manual setup, rely on passwords, and operate at the device level—making them easy to bypass. There's a critical need for intelligent, automated protection that works in real-time.

Real-Time Protection

Automated blocking without constant supervision

AI-Powered Intelligence

Computer vision-based age verification

Browser-Level Security

Works directly where access happens

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INNOVATION

How It Works & What Makes It Unique

01

Camera-Based Age Estimation

Deep learning CNN model analyzes facial features through device camera to estimate user age in real-time

02

Instant Access Control

If estimated age is below threshold or camera is disabled, restricted sites are automatically blocked with 404 page

03

Privacy-First Design

All inference performed locally on device—no data transmitted externally

04

Cross-Browser Compatibility

Works seamlessly across Chrome, Firefox, Opera, and other popular browsers

Rigorous Testing & Global Scalability

The system is trained on diverse datasets including IMDB-WIKI, UTKFace, and FairFace, evaluated through age estimation accuracy, mean absolute error (MAE), and blocking performance metrics. This first-of-its-kind browser-level biometric age gate can scale globally across schools, parental control systems, cyber cafes, and regulatory compliance frameworks.

Future enhancements: emotion detection, content severity scoring, and multilingual support to expand protection worldwide.

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